

Chapter 13: Vector Database Internals & HNSW Graph Math

Zero-Prerequisite Intuition: The "Galaxy Star Map" Metaphor What is a vector embedding, and why do we need special Vector Databases (Pinecone, Qdrant, Milvus, Chroma)? Imagine looking up at the night sky. Every star has 3D coordinates in space: (X, Y, Z) .

- Earth is at $(0, 0, 0)$.
- Mars is nearby at $(0.5, 0.2, 0.1)$.
- The Andromeda Galaxy is far away at $(2500000, 890000, 120000)$. Because Earth and Mars have coordinates that are close to each other, you know they are in the same neighborhood!

A **Vector Embedding Model** does the exact same thing for human thoughts and text. It reads a sentence and assigns it coordinates in a **1,536-dimensional space**:

- "The dog barked" $\rightarrow [0.24, -0.81, 0.12, \dots]$
- "A puppy made a sound" $\rightarrow [0.26, -0.79, 0.11, \dots]$ (Extremely close coordinates!)
- "Stock market interest rates rose" $\rightarrow [-0.91, 0.34, -0.88, \dots]$ (Far away coordinates!)

But here is the crisis: If you have **10,000,000 document vectors**, computing the exact distance between your search query and all 10 million vectors requires billions of floating-point operations. Your search takes **6 seconds**!

A **Vector Database** uses **Approximate Nearest Neighbors (ANN)** to search 10 million vectors in **less than 8 milliseconds**!

1. Distance Metrics: Cosine vs. Dot Product vs. Euclidean ($L2$)

How do we mathematically measure the "distance" between two thought vectors $\mathbf{u}$ and $\mathbf{v}$?

Mathematical Similarity Metrics

Euclidean ($L2$): Direct Straight-Line Distance
 $d(\mathbf{u}, \mathbf{v}) = \sqrt{(\sum (\mathbf{u}_i - \mathbf{v}_i)^2)}$

Dot Product: Magnitude $\times$ Angle
 $\mathbf{u} \cdot \mathbf{v} = \sum (\mathbf{u}_i \times \mathbf{v}_i)$

Cosine Similarity: Pure Angular Direction
 $\cos(\theta) = (\mathbf{u} \cdot \mathbf{v}) / (||\mathbf{u}|| \times ||\mathbf{v}||)$

- **Euclidean (L_2) Distance:** Measures the geometric straight-line physical distance. Sensitive to vector magnitude (document length).
- **Dot Product:** Fast to compute on modern CPU/GPU hardware using SIMD instructions.
- **Cosine Similarity:** Measures the angle θ between vectors, completely ignoring document length.

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

TIP

Production Optimization: If you normalize all embedding vectors to unit length ($\|\mathbf{u}\| = 1$) upon ingestion, **Cosine Similarity becomes mathematically identical to Dot Product!** You can replace expensive divisions and square roots with a single blazing-fast hardware vector dot product.

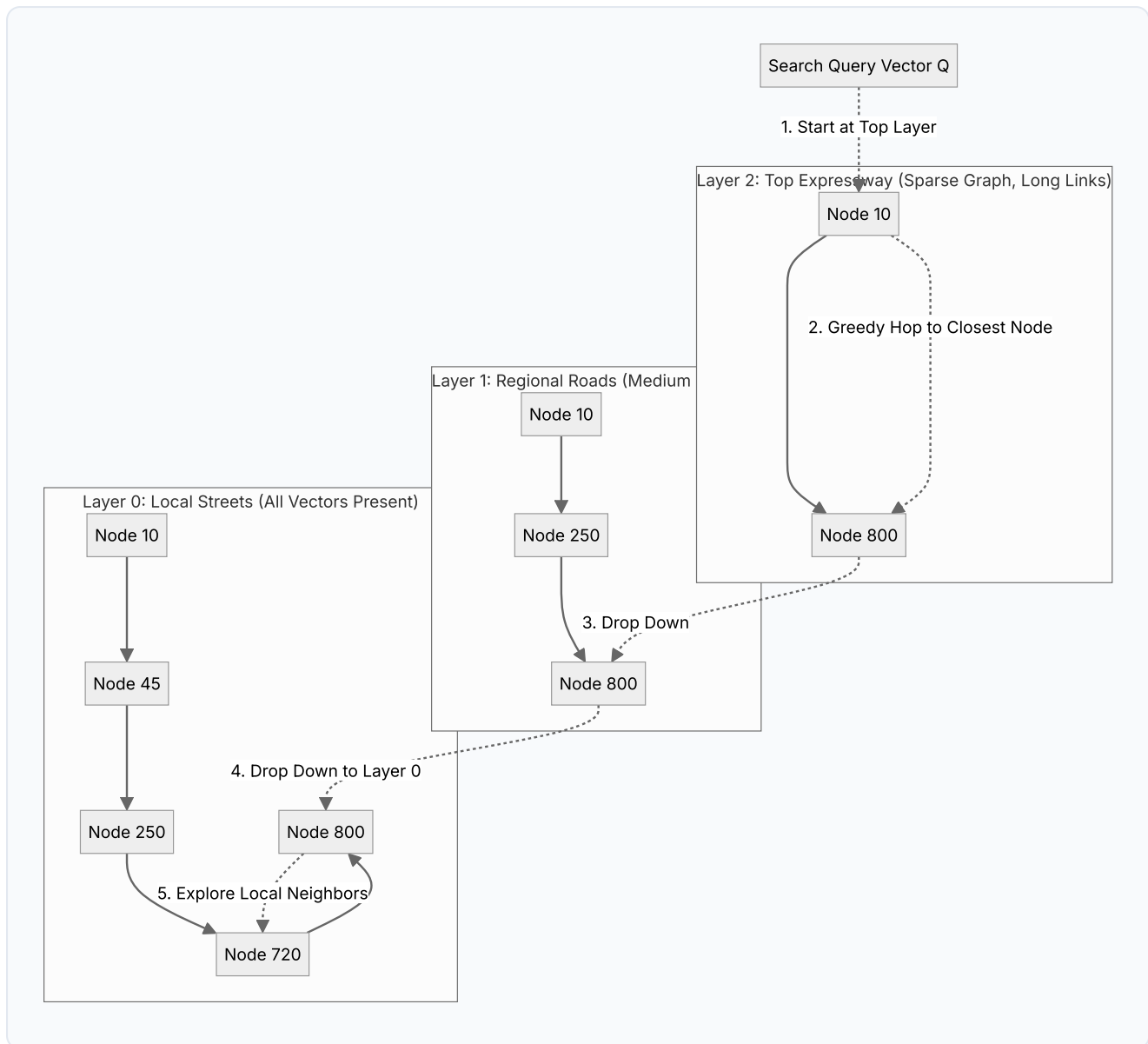
2. The HNSW Algorithm: Multi-Layer Skip-Graphs

The undisputed state-of-the-art algorithm for vector search is **Hierarchical Navigable Small World (HNSW)**.

The "Express Train vs. Local Walking" Metaphor

How do you travel from New York to a specific house in Los Angeles?

- You don't walk street-by-street across 3,000 miles (that is brute force $O(N)$).
- You take an **Express Airplane** across the continent (Layer 2).
- You land at LAX and take a **Regional Highway** to the neighborhood (Layer 1).
- You take a **Local Street** to the exact doorstep (Layer 0).



The Search Mechanics:

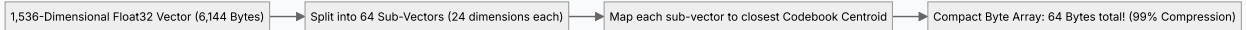
1. Start at the top layer entry point with greedy routing: jump to whichever neighboring node is closest to query vector Q .
 2. When no neighbor is closer to Q , drop down one layer and resume greedy routing from that node.
 3. At Layer 0 (the bottom layer containing all vectors), execute a **Beam Search** with size `efSearch` to extract the Top- K nearest neighbors.
- **Complexity:** Scales in logarithmic time: $O(\log N)$ rather than linear time $O(N)$! Searching 10 million vectors takes only ~20 graph hops!

3. Product Quantization (IVF-PQ): Compressing Vectors

Storing 10 million vectors of 1,536 floats in RAM requires:

$$10,000,000 \times 1536 \times 4 \text{ bytes} \approx \mathbf{61.4 \text{ Gigabytes of RAM!}}$$

RAM is expensive (\$100s/month). **Product Quantization (PQ)** compresses this by 95% down to **3 GB** using lossy vector quantization:



1. Divide the 1,536-dimensional vector into 64 sub-vectors of 24 dimensions each.
2. Run k -means clustering on each sub-space to generate 256 centroid prototypes.
3. Instead of storing 24 floats (96 bytes), store a single **8-bit integer (1 byte)** representing the centroid index!
4. Querying computes distance using pre-computed centroid lookup tables, speeding up distance computations by $10\times$.

4. Metadata Filtering: Pre-Filtering vs. Single-Stage Hybrid Search

In real applications, you rarely search vectors in isolation. You search: *"Find documents similar to 'billing refund' WHERE `user_id = 42` AND `created_year ≥ 2025`"*.

Filtering Strategy	How It Works	Danger / Failure Mode
Post-Filtering	Run HNSW vector search to find Top-100 nearest items, then discard items that don't match <code>user_id = 42</code> .	Zero Results Disaster: If only 1 out of 10,000 items belongs to user 42, the top-100 will contain 0 matching items, returning an empty list!
Pre-Filtering	Query SQL index first to find all IDs matching <code>user_id = 42</code> , then run brute force vector distance only on those matching IDs.	Inefficient if user 42 has 500,000 items (falls back to slow brute force).
Single-Stage Filtered HNSW (Qdrant / Milvus)	Navigates the HNSW graph while skipping graph nodes during beam search that fail the metadata bitset.	Optimal: Guarantees exact Top- K while preserving $O(\log N)$ graph search speeds.

4. Runnable Model: Memory Arithmetic and the Recall Cost of Searching Fewer Clusters

Why quantisation matters at a billion vectors

Takeaway: at scale the question is rarely "which index is fastest" but "what fits in memory". Quantised vectors for the first-pass search, plus the full vectors on disk for re-ranking the top candidates, is the standard compromise.

The IVF trade-off: probe more clusters, get more recall, pay more time

An IVF index partitions vectors into clusters and searches only the `nprobe` clusters closest to the query. This model builds one and measures recall against exact search.

The numbers vary with the random data, but the shape does not: **recall rises with `nprobe` and so does the work per query**. Production systems tune `nprobe` (or HNSW's `ef_search`) against a recall target measured on their own data, with a fixed latency budget; the ANN index never promises exact answers, so the recall measurement is part of the system.

Further Reading

- [HNSW paper](#)
- [Pinecone: HNSW explained](#)
- [FAISS wiki](#)

Check Yourself

Answer in your head or on paper first, then open each answer.

- ▶ 1. What does HNSW trade for speed?
- ▶ 2. What does `efSearch` control?
- ▶ 3. Why normalise vectors for cosine similarity?
- ▶ 4. How does product quantization save memory?